# Magnetide — Build Instructions v21-r2i

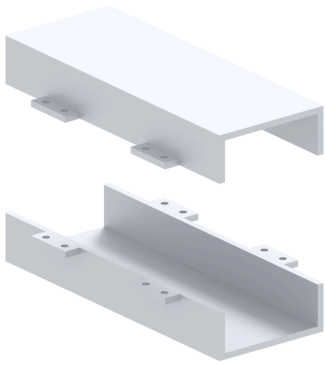
MHD seawater pump · 150 × 73 × 45 mm
Kelas 10.8, Kelompok 1 · Reynard Filbert Tang
Rev r2i · 18 parts · 7 steps

| 2× | 2× | 2× | 96× | 2× | 1+1 | 8× |
|---|---|---|---|---|---|---|
| YOKE | TRAY | CAP | MAGNET | ELECTRODE | ENDS | M3 BOLT |
| 3.0 mm steel · 150 long | 124 × 50.7 × 19.5 | 1 mm · Ø3 pegs | 20 × 10 × 5 N52 | graphite 134 × 14 × 2 | bellmouth · rear block | M3 × 14 + nut |

## 1 Fold the two yokes

Fold 90° **down** at ±28.5 on an **R3** die for the walls, then the two lugs **out** at ±44.06 on **R1.5**. One blank per half, mirror image.

> Check: 57 mm across the plate, 22.5 mm plate face to lug face. Two Ø3.4 holes per lug, 13 mm apart, at x 40 and x 110. R3 is not optional on 3 mm sheet — a sharp die splits the outside of the bend.

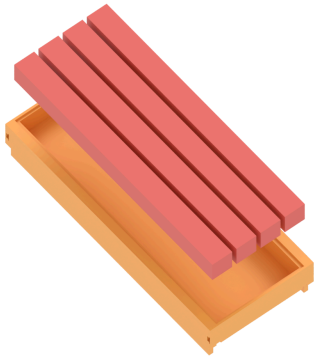


**Need:** 2 × clamshell-flat-3.0mm.dxf, folded · **Then:** deburr all edges

## 2 Load the magnets

48 tiles per tray: 6 along the bore, 4 across the width, 2 layers deep. They drop into the cavity and finish flush with the floor.

> **Compass check before the epoxy sets.** All four rows on a face point the same way; the other tray points the other way. One flipped row stalls a quarter of the bore.

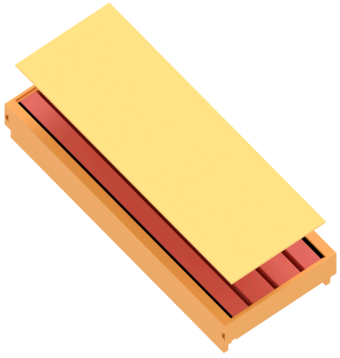


**Need:** 2 × tray, 96 × N52 tile, epoxy · **Drawn as 4 bars** = the 48 tiles of one tray · drop in, never press-fit

## 3 Cap it

Drop the 1 mm cap on the ledge. It sits on the magnets and finishes flush with the steel inner face.

> This face is the magnet's road into the yoke. A bead of epoxy at the rim only — **never** a puddle.

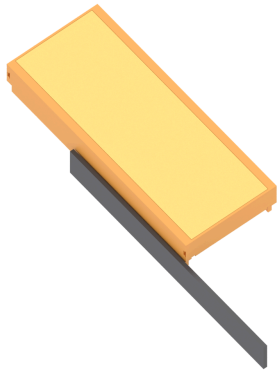


**Need:** 2 × cap · **Result:** stack height 19.5 mm per half

## 4 Slide in the electrodes

Each graphite plate slides into the 1 mm groove from the end. Push until it sits 5 mm inside the end pocket.

> Full 14 mm face exposed: 124 × 14 = 1736 mm² per plate. Keep the edges clean — a burr on graphite is a short waiting to happen.

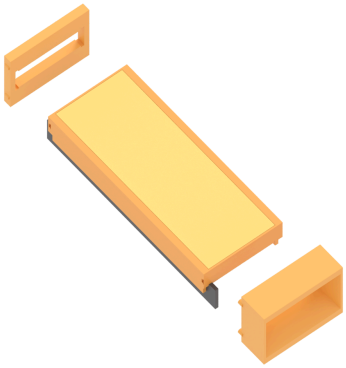


**Need:** 2 × electrode · **Fit:** 0.2 mm clearance in the groove

## 5 Bellmouth and rear block

Two Ø3 pegs each, into the blind holes in the tray end faces. Mouth in, straight end out.

> **Wire it now:** drill through the last 0.4 mm of the lead channel, feed the two wires out, epoxy the hole shut around them.

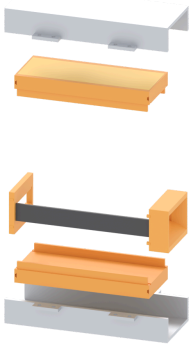


**Need:** 1 × bellmouth, 1 × rear block, epoxy · **Keep:** the outlet straight

## 6 Close the yoke

Lower tray into the bottom yoke, upper into the top. Bring the halves together so the lug faces meet bare — no paint, no gasket.

> This is a **stack clamp**: the tray rim lands on the steel plate, the mirrored tray butts it at mid-plane, the bolts close the sandwich.



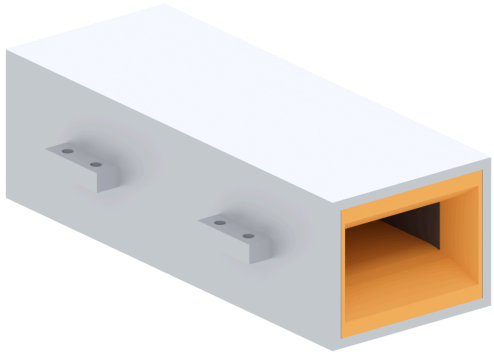
**Need:** 2 × assembled half · **Check:** split faces bare metal

## 7 Bolt it

Eight M3 × 14 through the lugs — **two per lug, 13 mm apart**. Tighten in a cross pattern, a little at a time, so the halves seat flat instead of rocking.

> 2 N·m gives roughly 8 kN across the four lug pairs. Run a nut and washer on the underside; thread never protrudes into the bore.

**Need:** 8 × M3 bolt, nut, washer · **Check:** no gap at any lug face



## Before you power it

- ☐ Both halves flush, 8 bolts seated, no gap at the lug faces
- ☐ Compass: every row on a face points the same way, the two faces oppose
- ☐ Tray floor unbroken over the whole magnet area — hold it up to a light
- ☐ Electrodes fully home, not shorting to steel through a scratch
- ☐ Lead exit drilled, wired, epoxied; bore clear down its whole 124 mm
- ☐ Fuse in the supply line; runs under 2 minutes, ventilated ($Cl_2$)

| If it does not fit | Nominal | Cause |
|---|---|---|
| tray rattles sideways | 0.15 / side | shim 0.3–0.5 mm closed-cell foam on the **outer** wall only |
| cap stands proud of the steel | 1.0 | epoxy puddle under the cap, or a magnet sitting proud |
| electrode will not enter | 2.0 in 2.2 | graphite burr, or the groove has epoxy in it |
| halves will not close | 0 gap | tray rim not seated on the plate — check the stack, not the bolts |